

Abstract

This project investigates the relationship between various personal lifestyle factors, and studying efficiency. Daily data will be collected on sleep schedule (sleeping and waking time), screen time (including most used apps and categories), caffeine intake (by mg), study location, total study duration, and self-rated sleep quality and study efficiency scores. Using this dataset, basic data analysis and statistical methods will be applied to examine any associations among these variables to examine the associations and test the hypothesis. Furthermore, supervised machine learning approach was used to model the relationships between lifestyle related factors and studying efficiency.

Motivation

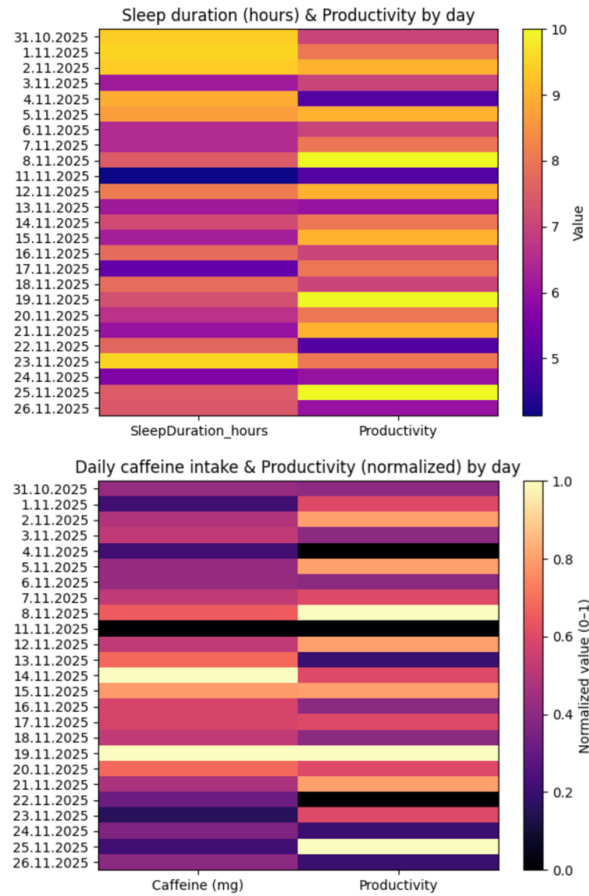
As a student, I always seek ways to improve my productivity and time management. I have always assumed that such lifestyle factors had huge influence on my efficiency, but I never had concrete data to prove it. In this project I aim to find how these factors affect my productivity. Ultimately, I hope to find an “optimal routine” to make the most out of my time in this world.

Data Sources

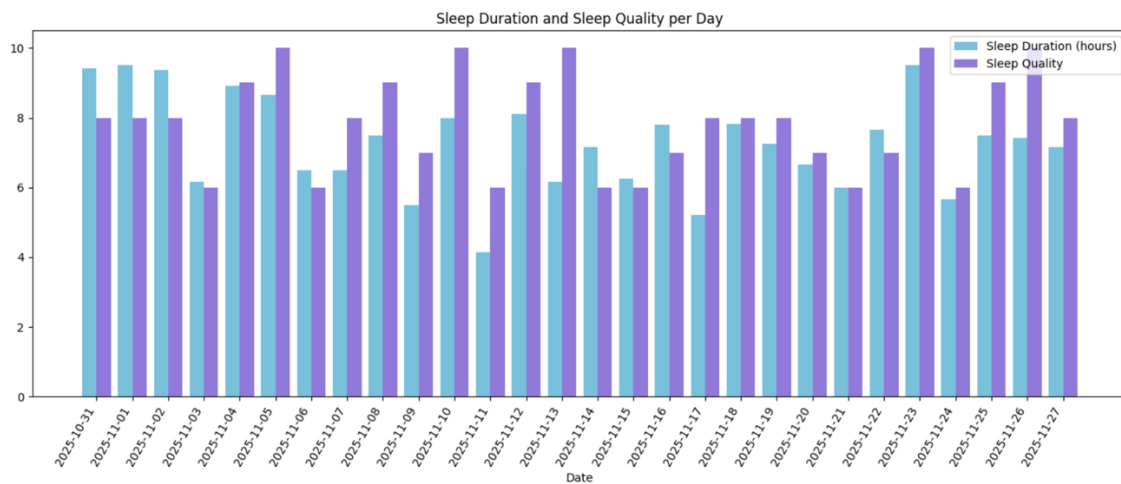
Daily data will be collected from multiple sources. Sleep-related information will include manually recorded sleeping and waking times, as well as a self-assessed sleep quality score ranging from 0 to 10 (0 being worst and 10 being best). Caffeine intake will be calculated based on the recordings of each caffeinated drink consumed in a they and how many mg of caffeine they have. Screen time data will be taken from my settings app which includes the total screen time of my phone, computer, and iPad. The screen time data also includes the top 3 categories and apps, which are also taken from the settings app. Study-related data will be collected by recording daily study duration using a stopwatch application. Study efficiency score will be self-assessed from 0 to 10 (0 being worst and 10 being best). Furthermore, places I study each day will be recorder manually.

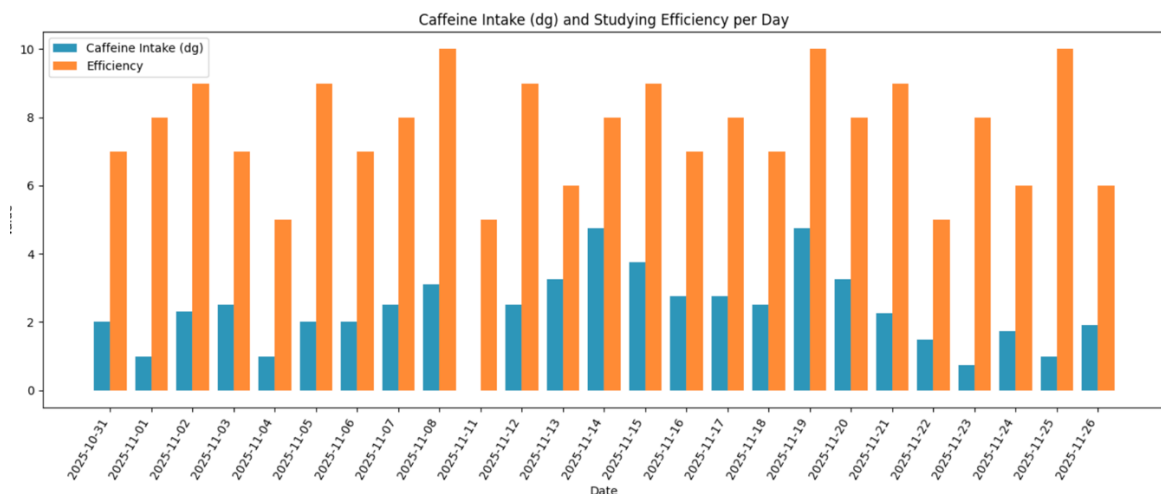
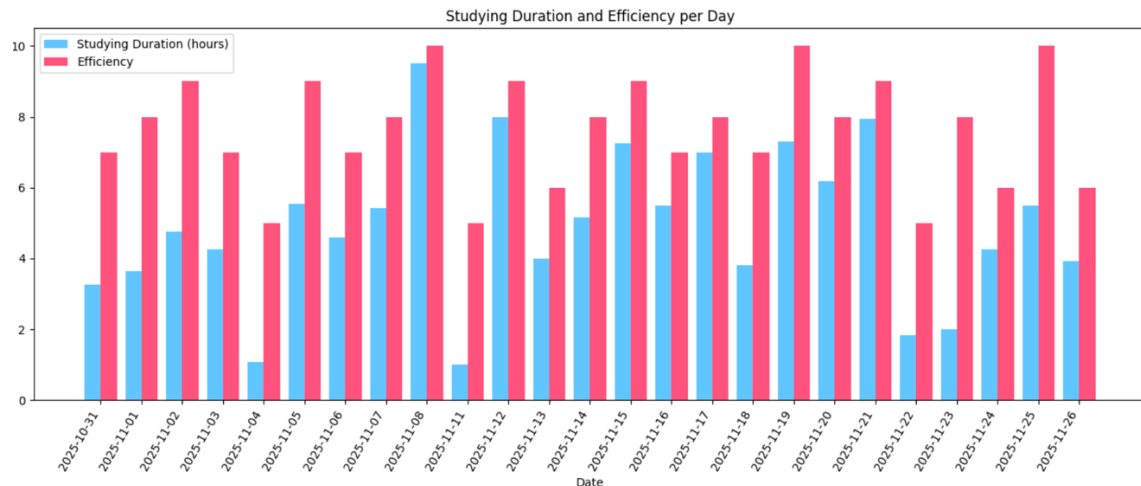
Exploratory Data Analysis (EDA)

The data analysis begins with loading the dataset and examining its structure. Daily sleep quality, caffeine intake, screen time, study time and productivity score are shown in a line chart. After that, the sleep duration-productivity score relationship and daily caffeine intake-productivity score relationship is visualized using a heatmap.



Lastly, grouped bar charts were used to visually examine the sleep duration-sleep quality, studying duration-efficiency, and caffeine intake-efficiency relationships.





Hypotheses & Hypothesis Testing

This project focuses on the following hypotheses:

Hypothesis H1: Sleeping between 7.5 to 9 hours per night is associated with high quality sleep and higher productivity scores.

Hypothesis H2: Studying at the Information Center results in higher productivity scores on average compared to any other location.

Hypothesis H3: Studying for more than 4 hours in a day results in a higher productivity score compared to studying 4 hours or less.

Hypothesis H4: Days with caffeine intake greater than 200 mg have higher productivity than days with 200 mg or less.

H1 (Sleep and Productivity) – Result:

At $\alpha = 0.05$, H1 is partially supported. Sleep duration is clearly related to sleep quality, but it does not show a significant relationship with productivity in this dataset.

Sleep duration $\leftrightarrow$ Sleep quality: significant, moderate positive relationship

Pearson $r = 0.543$, $p = 0.0028$; Spearman $r = 0.574$, $p = 0.0014$

Also, 7.5–9h sleepers reported higher sleep quality (8.67 vs 7.47, Welch $p = 0.0261$)

Sleep duration $\leftrightarrow$ Productivity: not significant

Pearson $r = 0.196$, $p = 0.3482$; Spearman $r = 0.153$, $p = 0.4646$

7.5–9h vs outside productivity difference is tiny and not significant (7.75 vs 7.59, Welch $p = 0.8427$)

Conclusion: Sleep duration (especially 7.5–9h) is associated with better sleep quality, but the data does not provide evidence that it increases productivity.

H2 (Studying Location) – Result:

At $\alpha = 0.05$, H2 is supported when “IC” is defined strictly as days where the study place is exactly "ic". Productivity on exact-IC days is significantly higher than on all other days.

Exact IC vs all other places (group comparison): significant

Welch t-test $p = 0.0052$, Mann–Whitney U $p = 0.0082$

Mean productivity: 8.55 (IC exact, $n = 11$) vs 6.93 (non-IC, $n = 14$)

Conclusion: Studying exactly at IC is associated with higher productivity in the dataset (statistically significant).

H3 (Study Duration) – Result:

At $\alpha = 0.05$, H3 is supported. There is a strong positive relationship between total study time and productivity, meaning that days with more study hours tend to have higher productivity scores.

Study hours $\leftrightarrow$ Productivity (correlation): significant, strong positive

Pearson $r = 0.796$, $p \approx 0.0000$

Spearman $r = 0.799$, $p \approx 0.0000$

Conclusion: Longer studying time is strongly associated with higher productivity in this dataset.

H4 (Caffeine Intake) – Result:

At $\alpha = 0.05$, H4 is supported when caffeine intake is split into > 200 mg vs ≤ 200 mg. Productivity is significantly higher on days with more than 200 mg caffeine.

>200 mg vs ≤ 200 mg (group comparison): significant

Welch t-test $p = 0.0448$, Mann–Whitney U $p = 0.0481$

Mean productivity: 8.21 (>200 mg, $n = 14$) vs 6.91 (≤ 200 mg, $n = 11$)

If you report the hypothesis in a directional way (“ >200 mg increases productivity”), the one-sided tests also support it:

Welch (one-sided) $p = 0.0224$

Mann–Whitney (greater) $p = 0.0240$

Conclusion: Days with caffeine > 200 mg are associated with higher productivity in the dataset.

Machine Learning

Supervised machine learning approach was used to model the relationship between lifestyle-related factors and studying efficiency. After preprocessing the dataset and converting all time-based features into numerical form, several regression algorithms with varying complexity levels were implemented and evaluated.

The dataset was repeatedly divided into training and test subsets to assess model generalization. Models were trained exclusively on the training data and evaluated on unseen test data using Root Mean Squared Error (RMSE) and the coefficient of determination (R^2) as performance metrics. To ensure fair comparison, identical data splits and evaluation criteria were applied across all models.

Models Evaluated

- Multiple Linear Regression
- Polynomial Regression (degree 2)
- k-Nearest Neighbors Regression
- Decision Tree Regression
- Support Vector Regression (SVR)

Machine Learning Findings

The Decision Tree model achieved the best predictive performance, the tuned Support Vector Regressor was second. Linear Regression showed moderate effectiveness, while the default SVR and kNN models performed relatively worse. The baseline predictor and Polynomial Regression exhibited the weakest performance. Overall, models capable of capturing non-linear relationships provided superior results.

	Model	RMSE	R2
0	Decision Tree	0.707107	0.850610
1	SVR (tuned)	0.879777	0.768742
2	Linear Regression	1.094204	0.642275
3	SVR (default)	1.195751	0.572797
4	kNN (k=5)	1.366957	0.441707
5	Baseline(mean)	1.832371	NaN
6	Poly Regression (deg2)	5.327307	-7.479450

Later, these tests were repeated with many random splits instead of a single split.

	Model	RMSE_mean	RMSE_std	R2_mean	R2_std
0	Linear	1.111912	0.221991	0.282467	0.495387
1	SVR(tuned)	1.195857	0.275666	0.164198	0.632175
2	SVR(default)	1.270873	0.256448	0.106397	0.505294
3	kNN(k=5)	1.272357	0.219530	0.104063	0.570586
4	Tree(depth=3)	1.323623	0.321759	-0.052368	0.772916
5	Poly2	4.385766	2.414555	-12.029108	16.889999

We can conclude that the decision tree achieved the best performance on a single train-test split, but repeated evaluation over many random splits reveal that its unstable. Linear regression achieves the lowest average RMSE and the most stable performance amongst splits, making it the most reliable overall. SVR benefits from hyperparameter tuning, but does not consistently outperform Linear Regression. Polynomial Regression performs the worst due to overfitting.

Limitations & Future Work

One limitation of this study was time, data was not collected for a long period of time. Collecting data in longer periods of time would result in more accurate and reliable results. Furthermore, repeating this project across a larger and more diverse group (eg. a group of students) could provide insights on common and different factors for efficiency. Finally, recording studying time not only as duration but also as what time of the day may also be beneficial for future work, as an important and relevant factor in efficiency.